



Artificial intelligence models for tooth-supported fixed and removable prosthodontics: A systematic review

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ABSTRACT

Statement of problem. Artificial intelligence applications are increasing in prosthodontics. Still, the current development and performance of artificial intelligence in prosthodontic applications has not yet been systematically documented and analyzed.

Purpose. The purpose of this systematic review was to assess the performance of the artificial intelligence models in prosthodontics for tooth shade selection, automation of restoration design, mapping the tooth preparation finishing line, optimizing the manufacturing casting, predicting facial changes in patients with removable prostheses, and designing removable partial dentures.

Material and methods. An electronic systematic review was performed in MEDLINE/PubMed, EMBASE, Web of Science, Cochrane, and Scopus. A manual search was also conducted. Studies with artificial intelligence models were selected based on 6 criteria: tooth shade selection, automated fabrication of dental restorations, mapping the finishing line of tooth preparations, optimizing the manufacturing casting process, predicting facial changes in patients with removable prostheses, and designing removable partial dentures. Two investigators independently evaluated the quality assessment of the studies by applying the Joanna Briggs Institute Critical Appraisal Checklist for Quasi-Experimental Studies (nonrandomized experimental studies). A third investigator was consulted to resolve lack of consensus.

Results. A total of 36 articles were reviewed and classified into 6 groups based on the application of the artificial intelligence model. One article reported on the development of an artificial intelligence model for tooth shade selection, reporting better shade matching than with conventional visual selection; 14 articles reported on the feasibility of automated design of dental restorations using different artificial intelligence models; 1 artificial intelligence model was able to mark the margin line without manual interaction with an average accuracy ranging from 90.6% to 97.4%; 2 investigations developed artificial intelligence algorithms for optimizing the manufacturing casting process, reporting an improvement of the design process, minimizing the porosity on the cast metal, and reducing the overall manufacturing time; 1 study proposed an artificial intelligence model that was able to predict facial changes in patients using removable prostheses; and 17 investigations that developed clinical decision support, expert systems for designing removable partial dentures for clinicians and educational purposes, computer-aided learning with video interactive programs for student learning, and automated removable partial denture design.

Conclusions. Artificial intelligence models have shown the potential for providing a reliable diagnostic tool for tooth shade selection, automated restoration design, mapping the preparation finishing line, optimizing the manufacturing casting, predicting facial changes in patients with removable prostheses, and designing removable partial dentures, but they are still in development. Additional studies are needed to further develop and assess their clinical performance. (*J Prosthet Dent* 2023;129:276-92)

Artificial intelligence (AI) models have been applied in different dental specialties.¹⁻⁴ In restorative dentistry, AI has been applied to improve dental caries diagnosis by using periapical and bitewing radiographs,⁵⁻⁷ predict the

failure of a dental restoration,^{8,9} and diagnose vertical tooth fracture by using periapical radiographs or cone beam computed tomography data.^{10,11} In implant dentistry, AI has been used to identify implant type from

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Clinical Implications

Artificial intelligence models are valid tools to improve tooth shade selection, automated restoration design, map the preparation finishing line, optimize the manufacturing casting procedures, predict facial changes in patients with removable prostheses, and designs of removable partial dentures.

periapical and panoramic radiographs¹²⁻¹⁴ and in periodontics to enhance the diagnosis of periodontal disease.^{15,16} In endodontics, AI has been used to diagnose periapical lesions and locate the apical foramen.^{17,18}

AI models have been applied for different prosthodontics applications such as mapping the finishing line of tooth preparations or assisting in tooth anatomy selection for the automated design of dental restorations by using computer-aided design (CAD) methods.^{19,20} In addition, AI models have been developed to estimate the optimal parameters for the successful casting of a metal framework²¹ or to perform tooth shade selection and provide a recommended porcelain selection to accomplish shade matching.²²

Tooth- and implant-supported removable partial dentures (RPDs) provide a minimally invasive and cost-effective treatment option for replacing missing teeth.²³⁻²⁵ Designing the different elements of an RPD constitutes an important step in the prosthesis fabrication, but agreement on the optimal design is lacking.²⁶⁻²⁸ Similarly, AI algorithms have been developed to assist in the design of RPDs.

The clinical applicability and technology maturation stage of AI applications in prosthodontics remains uncertain; hence, an assessment of the development, performance, and limitations of the AI models for prosthodontic purposes is needed. This systematic review aimed to evaluate the performance of AI models for prosthodontic applications, namely shade matching, automated tooth anatomy design, locating the finishing line of tooth preparations, simulating the manufacturing casting process, predicting facial changes in patients with removable prostheses, and designing RPDs.

MATERIAL AND METHODS

The problem or population, intervention, comparison, outcome (PICO) question was formulated with the population comprising fixed and removable prostheses; the intervention including AI learning; the comparison was determined to be nonapplicable; and the outcome the performance of the AI model for shade matching, automated tooth anatomy design, locating the finishing line of tooth preparations, simulating the manufacturing

Table 1. Boolean search strategy used on 5 databases explored

Database	MeSH Terms and Search Terms
MEDLINE/PubMed	("Dentures"[MeSH] OR "Denture, partial"[MeSH] OR "Denture, partial, removable"[MeSH] OR "Dental prosthesis"[MeSH] OR "Denture Precision Attachment"[MeSH] OR "Denture, Partial, Immediate"[MeSH] OR "removable partial denture" OR "complete denture" OR "Crowns"[MeSH] OR "Dental prosthesis design"[MeSH] OR "Dental restorations"[MeSH] OR "tooth, artificial"[MeSH] OR "dental marginal adaptation"[MeSH] OR "finishing line" OR "fixed prosthodontics" OR "dental casting technique"[MeSH] OR "Color selection" OR "Shade selection") AND ("Artificial intelligence"[MeSH] OR "Computational Intelligence" OR "Machine Intelligence" OR "Computer Reasoning" OR "AI-based" OR "Computer Vision Systems" OR "Knowledge Acquisition" OR "Knowledge Representation" OR "Machine learning"[MeSH] OR "Deep learning"[MeSH] OR "Supervised machine learning"[MeSH] OR "Unsupervised Machine Learning"[MeSH] OR "Expert systems"[MeSH] OR "Fuzzy Logic"[MeSH] OR "Natural Language Processing"[MeSH] OR "Neural Networks, Computer"[MeSH])
EMBASE, Web of Science, Cochrane, and Scopus	("Dentures" OR "Denture, partial" OR "Denture, partial, removable" OR "Dental prosthesis" OR "Denture Precision Attachment" OR "Denture, Partial, Immediate" OR "removable partial denture" OR "complete denture" OR "Crowns" OR "Dental prosthesis design" OR "Dental restorations" OR "tooth, artificial" OR "dental marginal adaptation" OR "finishing line" OR "fixed prosthodontics" OR "dental casting technique" OR "Color selection" OR "Shade selection") AND ("Artificial intelligence" OR "Computational Intelligence" OR "Machine Intelligence" OR "Computer Reasoning" OR "AI-based" OR "Computer Vision Systems" OR "Knowledge Acquisition" OR "Knowledge Representation" OR "Machine learning" OR "Deep learning" OR "Supervised machine learning" OR "Unsupervised Machine Learning" OR "Expert systems" OR "Fuzzy Logic" OR "Natural Language Processing" OR "Neural Networks, Computer") NOT MEDLINE

casting process, predicting facial changes in patients with removable prostheses, and designing RPDs. Five databases were selected without any date restriction: MEDLINE/PubMed, EMBASE, Web of Science, Cochrane, and Scopus. A manual search was also conducted (Table 1). Data search included articles published between January 1980 and February 2021.

All titles and abstracts were first assessed for the following inclusive criteria, which included clinical or in vitro studies that evaluated the performance of the AI models in assisting shade matching, automated tooth anatomy design, locating the finishing line of tooth preparations, simulating the manufacturing casting process, predicting facial changes in patients with removable prostheses, and designing RPDs. This systematic review conformed to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.²⁹

After evaluating the full text of the articles as per the previously defined inclusive criteria, AI studies related to other disciplines, AI models developed for other dental

disciplines such as endodontics, periodontics, maxillo-facial surgery, pediatric dentistry, and orthodontics, studies developing AI models for tooth segmentation, review articles of AI models, AI model not described, studies related to robotics in dentistry, and radiographic image enhancement investigations were considered ineligible.

Two calibrated investigators (M.R.L., M.G.P.) collected the data from the selected articles into structured tables. Discrepancies were resolved by consensus with a third examiner (V.K.). The same 2 investigators independently evaluated the quality assessment of the studies by applying the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Quasi-Experimental Studies (nonrandomized experimental studies) (Table 2).³⁰ The third examiner (V.K.) was consulted to resolve lack of consensus.

RESULTS

The AI models found among the included studies comprise knowledge-based and expert systems, classical AI methods such as statistical, computer vision, and case-based reasoning models, and artificial neural networks (ANN) such as backpropagation neural networks (BPNN) and deep convolutional neural networks (Table 3). The included publications dated from 1985 to 2021.

The search strategies yielded 250 articles. After evaluating the titles and abstracts, 45 articles were identified, 9 of which were excluded after full-text revision (Fig. 1). The 36 selected articles were classified into 6 groups based on the application of the AI model: tooth shade selection,²² automated tooth anatomy design (Table 4),^{20,31-37,45-48,60,61} locating the finishing line of tooth preparations,¹⁹ simulating the manufacturing casting process (Table 5),^{21,50} predicting facial changes in patients with removable prostheses,⁶² and designing RPDs (Table 6).^{38-44,49,51-59}

With respect to the selection of articles through reviewing their titles and abstracts, there was significant agreement between the 2 reviewers for the articles that were selected (Kappa=0.96, *P*<.001) and the articles that were not selected (Kappa=0.97, *P*<.001). With respect to the selection of articles through reviewing the full text, there was a significant agreement between the 2 reviewers for the articles that were selected (Kappa=1, *P*<.001) and the articles that were not selected (Kappa=0.92, *P*<.001).

Because no specific in vitro study quality assessment tool has been developed, questions 2 and 6 of the Joanna Briggs Institute Critical Appraisal Checklist for Quasi-Experimental analysis were not applicable in this systematic review. Questions 3, 5, and 7 were not applicable for any of the included studies. The Joanna Briggs Institute Critical Appraisal Checklist for Quasi-Experimental results

Table 2. Joanna Briggs Institute Critical Appraisal Checklist for Quasi-Experimental Studies (nonrandomized experimental studies)

Question	Answer
1 Is it clear in the study what is the 'cause' and what is the 'effect' (i.e., there is no confusion about which variable comes first)?	Yes, No, Unclear, or Not applicable
2 Were the participants included in any similar comparisons?	
3 Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	
4 Was there a control group?	
5 Were there multiple measurements of the outcome both before and after the intervention/exposure?	
6 Was follow up complete and, if not, were differences between groups in terms of their follow up adequately described and analyzed?	
7 Were the outcomes of participants included in any comparisons measured in the same way?	
8 Were outcomes measured in a reliable way?	
9 Was appropriate statistical analysis used?	

showed a 100% low risk of bias in all included articles for question 1. For question 4, all the studies obtained a low risk of bias except for 13 studies^{32,38,39,50-54,56-59} that obtained a high risk of bias. For questions 8 and 9, all the investigations attained a low risk of bias except for Mackert⁴⁴ that showed a high risk of bias (Fig. 2).

DISCUSSION

The included articles were published between 1985 and 2021. Most of the studies published between 1985 and 1996 and included in the present review analyzed the AI models for RPD design. However, between 1997 and 2013, most of the studies published developed AI models for automated tooth anatomy design. Finally, since 2015, studies have developed AI models for broader prosthodontic applications.

Only 1 included study developed a BPNN model to perform tooth shade selection and provide a recommended porcelain instructional recipe to match the tooth color.²² The training database consisted of metal-ceramic disks of 3 different shades (A1, A2, and A3), and the disk color was measured with a spectroradiometer. After training and testing the AI model, the authors compared the shade selection of 10 central incisors by 10 volunteers with a commercial dental shade guide, the direct shade selection obtained by consensus of 3 experienced prosthodontists, and a computer color-matching system (spectroradiometer) based on an improved BPNN model. The authors concluded that the AI model had better shade matching capabilities than the direct visual assessment. However, how the basis or ground truth for the natural teeth color was obtained is unclear. The recommended porcelain formula was limited to a specific conventional ceramic system which consisted of a powder-liquid mixture and a successive

Table 3. Artificial intelligence models used in articles included in systematic review

Knowledge-Based and Expert Systems	Classical Methods	Artificial Neural Networks
Knowledge-based systems: computer programs that reason using knowledge bases to solve problems. ³¹⁻⁴⁴	Statistical models: mathematical models that use statistical assumptions typically for data generation. ⁴⁵⁻⁴⁸	Artificial neural networks or neural networks: based on collection of connected units or nodes called artificial neurons. Artificial neuron that receives signal then processes it and can signal neurons connected to it. Connections called edges. Typically, neurons aggregated into layers.
Expert systems: type of knowledge-based system that takes decisions based on set of expert heuristics ^{21,50-59}	Computer vision methods: methods that help computers gain understanding from digital images. ^{20,60,61}	• Backpropagation NN¹⁻² • Deep convolutional neural networks^{19,49}
	Case-based reasoning: Manages cases (past experiences) to solve new problems. ^{43,44}	

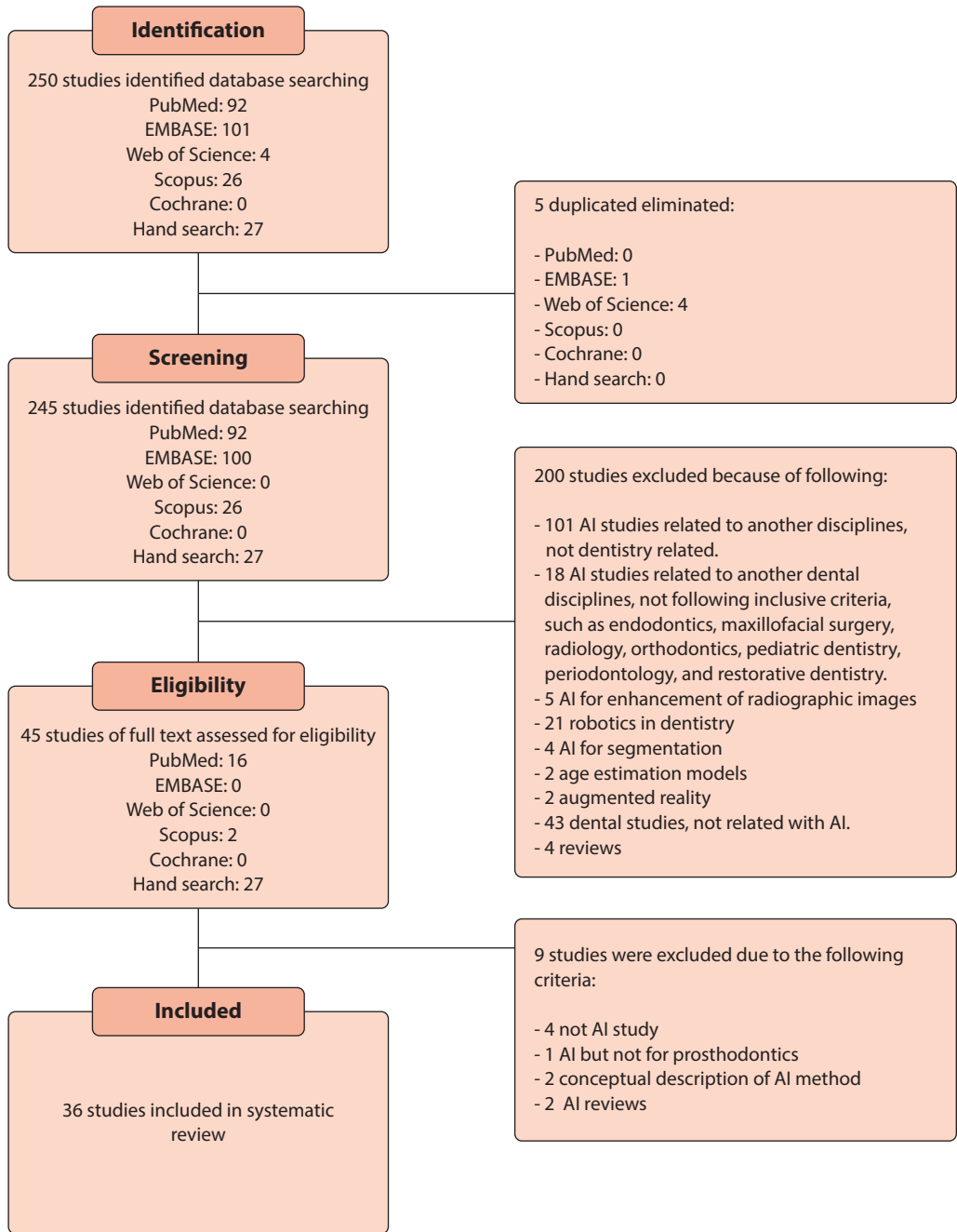


Figure 1. PRISMA flow diagram with information through phases of the study selection.

Table 4. Study characteristics of AI models for automated dental restoration designs

Reference	AI Model	Application	Implementations	Performance
Gürke, 1997 ³¹	Knowledge-based system (Geometrically deformable model (GDM))	Automation of tooth restoration designs	Data: Polyclinic and Restorative Dentistry and Periodontology in Munich provided 3D scans Tooth preparations performed. Method: Generalized concept of GDM by introducing free function assignment in control points. Automatically detect occlusal surface of inlay preparation during adaptation process of model. Basic steps as follows: - Triangular mesh placed on structure to be restored and deformed till it matches object's shape. - Deformation occurs by vertex translation and introduction of new vertices - Shape resemblance by choosing vertex locations to optimize function whose terms describe features of object. - Stochastic optimization used to find first approximation of ideal vertex locations. - Fast gradient method used to find exact location. - Occlusal surface then detected automatically when object shape and GDM sufficiently similar.	Mean distance error calculated in control points of GDM ranged from 1.20 to 6.48 μm .
Paulus et al, ²⁰ 1999	Computer vision methods (feature detection, image warping)	Automation of tooth restoration designs	Three applications for reconstruction of surface information introduced: - First aimed at building database of normalized depth images of posterior teeth and then extracting characteristic features from these images. - Second digitally reconstructed given occlusal surface of posterior tooth with prepared cavity from intact model tooth from database. This could then be used for automatic milling of dental prosthesis. - Third, laser scanner-scanned and 3D-copied hand-made interim wax inlay or crown into different material (such as, ceramic). Results converted to format required by computer-integrated manufacturing for automatic milling. Database input: Digitized (CEREC sensor) 60 intact maxillary first molars from 40 patients (15 to 30 years old).	Results: First application: Database created. To judge quality of automatic detection of cusp tips, 20 range images evaluated manually by dentist, twice. Average distance between automatic and human evaluation approx. 0.17 mm, with 70% patients having <0.1 mm distance between marked tip positions. Second application: dentist manually blackened 10 images to simulate cavities of 3 different sizes. Software program used to reconstruct damaged areas. Afterward, areas compared with original occlusal surfaces. Mean height difference between original and computed surfaces in range of 0.2 to 1.0 mm. Third application: System tested with several objects in addition to teeth. Usually, 10 images used for construction of complete 3D models. Library of digitized intact tooth surfaces can be built by image processing methods that extract features from surfaces; most suited model surface for given prepared tooth can be selected by human operator.
Blanz et al, ⁴⁵ 2004	Statistical model (using singular value decomposition)	Tooth CAD modeling	Data: Digitized stone casts of 166 intact first maxillary molars (from 6 to 9 y of age). Method: - 3D scans first converted into Morphable Models. - Missing information recovered using correlation between vector components stored in Models. - Statistically motivated regularization used to improve stability of results. - Optimum computed in direct, non-iterative way at interactive rates using SVD. Algorithm robust with respect to unknown orientation. Assessment of software program result: On 66 randomly selected scans, 43% of occlusal surface removed and reconstructed by software program (original geometry considered ground truth).	Average error between x and y points of reconstructed surfaces and true surface 143 μm on missing regions and 85 μm on remaining model. The root-mean-square errors between 3D positions of closest neighbors 137 μm on missing regions and 95 μm on remaining model.

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Table 4. (Continued) Study characteristics of AI models for automated dental restoration designs

Reference	AI Model	Application	Implementations	Performance
Mehl and Blanz, ³² 2005	Knowledge-based system (biogeneric tooth model)	Automation of tooth restoration designs	Input data: 200 to 300 tooth surfaces without caries from individuals from 6 to 20 y of age measured 3D for each type of tooth. Biogeneric tooth model creation: - First, common features and correlations of features sought by learning algorithms for each tooth type separately. This lead at first to reconstruction of "average tooth" which had common structures and available as data record in same resolution as measured data records. - After this analysis, deviations of library teeth from average tooth determined in a second step by PCA, in which most frequently occurring deviations stored in first principal components and less frequent deviations in last principal components. - Last step: reconstruction of occlusal surfaces adapting biogeneric tooth model to residual tooth material, antagonist, and neighboring tooth. Assessment of software program result: 40 simulated inlay cavities.	Morphology of library teeth described about 80% accurately with 20 principal components. Majority of naturally occurring tooth surfaces can already be synthesized by variation of first 20 principal components or parameters. Results showed that fully automatic reconstruction possible except for 4 situations. In majority of the situations, no further interactive improvements would have been necessary.
Mehl et al, ³³ 2005	Knowledge-based system (biogeneric tooth model)	Automation of tooth restoration designs	Input data: 170 mandibular first molars (based on set of impressions of carious-free and intact tooth surfaces from individuals 6 to 9 years old, digitized using scanning device). Biogeneric tooth model creation: - Average tooth computed capturing common features of molar's surface quantitatively. Crucial step to establish dense point-to-point correspondence between all teeth. Algorithm did not involve any prior knowledge about teeth. - Occlusal surface reconstruction: In second step, analysis of the PCA. Repeated for 3 different reference teeth, procedure yielded average teeth nearly independent of reference. Assessment of software program result: 5 inlay cavity simulations, visual evaluation with original surfaces.	41 principal components of tooth morphology. First 5 represented 52% of total variance, first 10 principal components 72%, and first 20 principal components 83%. Reconstruction performed from nothing more than information of 14 selected 3D points belonging to original surface. Visual inspection showed that structures and features of calculated surfaces similar to those of original surfaces. In none of 5 situations visual inspection revealed any atypical tooth structures, deformations, or errors.
Richter and Mehl, ³⁴ 2006	Knowledge-based system (biogeneric tooth model)	Automation of tooth restoration designs	Data input: 9 intact molar surfaces not contained in tooth library (6 maxillary and 3 mandibular molars) 3D measured. In each situation, simulation of 3 cavities of different size by drawing preparation boundaries on monitor, making possible to compare calculated occlusal surface with original occlusal surface. Reconstructed surface of 27 inlay cavities using biogeneric tooth model in DentVisual 2005 program. Four reference points determined, reconstructions generated by combining different variables: plausibility factor, number of principal components, and number of iterations. Assessment of software program result: visual evaluation performed by authors and another expert (contour, position of cusps, cusp shape, fissure pattern, and fissure depth as criteria).	Reconstruction deviations from original tooth surface on average around 150 μm. With aid of biogeneric tooth model, possibility of reconstructing occlusal surfaces of dental restorations by CAD-CAM processes fully automatically, taking account of morphological and functional criteria.
Sporring and Jensen, ³⁶ 2007	Statistical model (using PCA, Bayesian framework)	Automation of tooth restoration designs	Training dataset for each tooth: 12 3D images. 3D statistical multiobject shape model used for restoration system, built using 3D scan of dental casts. This system extension of the PCA, which included elasticity term for covariance matrix and collision avoidance for antagonist teeth. Improved generalizability for reconstruction process.	Reconstruction performed well in terms of missing data; collisions minimized for improved occlusion.

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Table 4. (Continued) Study characteristics of AI models for automated dental restoration designs

Reference	AI Model	Application	Implementations	Performance
Sporring and Jensen, ³⁷ 2008	Statistical model (using PCA, Bayes Method)	Automation of tooth restoration designs	For each tooth, 12 digitized casts with articulated antagonist obtained by using laboratory scanner (3Shape). Extension of PCA which included both elasticity term for co-variance matrix and collision avoidance for antagonist teeth, for better generalizability, conditions given under which elasticity is Bayes estimate.	Reconstruction generalized well in terms of missing data, minimizing collisions for improved occlusion. Preliminary clinical evaluation indicated that resulting casts visualized by variational implicit surfaces more natural looking than standard reconstructions.
Yuan et al, ⁶⁰ 2010	Computer vision methods (feature region extraction, segmentation)	Tooth CAD modeling	Tooth modeling sequence: - Digitized dental cast by using 3D scanner (accuracy <10 μm). - Analyze, select, and remove fusion regions between adjacent teeth - Restore shape of single tooth - Analyze and select blending region between teeth and soft tissues. - Extract segmentation boundary and separate tooth from 3D dental model. Proposed model compared with triangulation quality and efficiency with method proposed by Barequet and Sharir.	Average triangulation and restoration efficiency reached 20 000 V/s and 4000 V/s, respectively by using method proposed in this study. Vertex number of restoration surface patch usually 1/30~1/40 of original 3D dental cast. Whole shape modeling procedure of 3D dental cast required between 2 to 3 min.
Ender et al, ³⁵ 2011	Knowledge-based system (biogeneric tooth model)	Automatization of tooth restoration designs	Objective: evaluate potential of biogeneric tooth model for generating well-matched and adjusted CAD-CAM fabricated partial-coverage crowns. Study: 12 patients (17 to 34 y of age), diagnostic casts mounted on articulator. 24 onlay preparations completed on diagnostic casts by 3 dentists (12 replacing 1 cusp and 12 replacing 3 cusps). Preparations digitized using IOS (CEREC 3D). Each restoration designed using 2 CAD program: conventional (CEREC 3C CAD, v2.80) and based on biogeneric model (Bio.CAD) Assessment of software program result: visual evaluation by 10 experienced dentists and subtractive manufacturing of each restoration 3 times.	No significant differences appeared between 1- and 3-cusp restorations. Criteria "fissure relief" and "cusp shape" revealed highly significant preference for morphology created with Biog.CAD with 225 out of 240 scores for fissure relief and 176 out of 240 scores for cusp shape. Criteria "overall morphology" rated better for biogeneric tooth model. Criteria "distribution of occlusal contacts" rated "equal" with both software programs.
Zheng et al, ⁶¹ 2011	Computer vision methods (active contour models, radial basis function)	Automatization of tooth restoration designs	3D morphing process for occlusal surface reconstruction performed in 3 steps: - Initial tooth modeling through extracting cavity contour and loading standard tooth from tooth database, creating initial prosthetic tooth model. - Feature mapping occurred by identifying few corresponding feature points on standard tooth and tooth preparation, with establishment of one-to-one spatial relation of feature points. - Surface deformation process whereby radial basis function used as mapping function to define spatial relation of feature points in 2 teeth and interpolate intermediate feature positions, thus, deformation of standard tooth surface - Finally, by merging deformed standard tooth and tooth preparation and performing Boolean intersection, completion of missing occlusal surface reconstruction. Assessment of software program result: Inlay and crown occlusal surface reconstruction from scanned cast provided by school of Stomatology of Tianjin Medical University.	Reconstructions completed, but evaluation not provided. Qualitative difference between different methods showed that 3D morphing method introduced in study was low labor intensive, required short time without manual intervention, had straightforward algorithm complexity, and low cost. (Comparison between Free form deformation, Volker Blanz method, ViRus system, CEREC system, 3D morphing in this study)
Litzenburger et al, ³⁶ 2013	Knowledge-based system (biogeneric tooth model)	Automatization of tooth restoration designs	5 participants (mean age 25 y of age). Diagnostics casts digitized with preparations for inlays and onlays. For this, 1 molar tooth randomly chosen in each individual. For each preparation, at least one cusp untreated. Tooth preparations digitized by using IOS (CEREC 3D) and dental restorations designed by biogeneric CAD software program. 5 dental technicians (at least 7 y of	Mean shape similarity value of reconstructions made by technicians was $310.2 \pm 78.8 \mu$ m and CAD reconstructions (biogeneric models) were $222.0 \pm 47.7 \mu$ m. In 22 of 25 situations, reconstruction made by CAD software closer to original than reconstructions made by technicians. CAD reconstructions significantly more similar to original morphology than

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Table 4. (Continued) Study characteristics of AI models for automated dental restoration designs

Reference	AI Model	Application	Implementations	Performance
			experience) waxed restorations. All restoration designs superimposed, and shape similarity value calculated between program and technicians designs with original occlusal geometry.	conventional reconstructions made by dental technicians 1 and 3. However, technician numbers 2, 4, and 5 showed no significant difference compared with CAD group.
Kollmuss et al, ³⁷ 2013	Knowledge-based system (biogeneric tooth model)	Automatization of tooth restoration designs	Objective: mathematically compare and assess morphological agreement between original morphology and CAD-reconstructed, waxed and CAM partial coverage crowns. Study: 39 participants with 39 intact first molars from School of Dentistry, University of Munich. 39 crown preparations on diagnostic mounted casts by 39 dental students. Preparations scanned with IOS (CEREC BlueCam), 3 groups: conventional waxing of restoration made by 30-year experience dental technician, virtual design of restoration using biogeneric software program, and digitized milled restoration designed using biogeneric program. Discrepancies evaluated using original morphologies.	The mean difference between the natural tooth surface and the biogeneric reconstruction was 184 ±36 µm (volume/area) and 187 ±41 µm (z difference). Mean difference between natural tooth surface and waxing was 263 ±40 µm (volume/area) and 269 ±45 µm (z difference). Finally, mean difference between natural surface and milled restoration 182 ±40 µm (volume/area) and 184 ±41 µm (z difference). CAD software program allowed closer reconstruction of teeth than waxings even without cusps. Milling device precise enough to transfer CAD model to definitive restoration.
Zhang et al, ⁴⁸ 2017	Statistical Model (probabilistic point registration)	Tooth CAD modeling in partially edentulous dentition	2 digital dental casts employed by using 3D scanner (accuracy <0.1 mm) articulated in maximum intercuspation. Generated sequence of partially edentulous models by extracting 1 to 3 different teeth from maxillary and mandibular dental cast on each side. Total of 72 partially edentulous casts. Process: Dental segmentation that separates each tooth and partially edentulous position, detection of edentulous spaces, followed by dental reconstruction with symmetrical teeth (DRST), which uses symmetrical property of dentition (rigid transformation between 2 symmetrical teeth existing on left and right dental cast estimated through probabilistic point registration by matching 2 shapes). Assessment of software program result: reconstruction of 72 models, occlusion analysis (original occlusion considered ground truth, disarticulated and then rearticulated by DRST software program), and comparison with 3 real partially edentulous casts.	72 models successfully reconstructed by using DRST software program. Reconstructed occlusion: most of absolute errors less than 0.2 mm. Their maximum absolute mean error 0.38 mm.

3D, 3-dimensional; AI, Artificial intelligence, CAD, computer-aided design; CAM, computer-aided manufacturing; GDM, geometrically deformable model; IOS, intraoral scanner; PCA, principal component analysis; SVD, singular value decomposition.

layer of different ceramic colors, which might be difficult for a dental laboratory technician to replicate accurately. Additional studies might consider standardized ceramic blocks for milled dental restorations, which may facilitate the replication of the color recommended by the AI model.

A total of 11 included articles assessed the feasibility of AI models for the automatization of dental restorations, aiming to assist in CAD design procedures and reduce manufacturing time. A generalized challenge is to reconstruct or estimate the anatomy of dental surfaces, including the occlusal surface, even if there is incomplete input data in the images, such as onlay or inlay preparations, which must reproduce the original geometry as closely as possible. The objective would be for the AI software program to provide a design of the dental restoration which can be used for subtractive or additive manufacturing methods.

The articles reviewed that evaluated AI models for automatic CAD modeling or designing dental restorations^{20,31-37,45-48,60,61} can be divided into 2 main groups, namely investigations developing algorithms for such application^{20,32-34,45-48,60,61} and studies that evaluated commercially available dental software programs³⁵⁻³⁷ at the time of the publication. All the studies developing AI models were able to construct a tooth library from digitized natural dentition, from which features of the occlusal surfaces of the teeth were extracted to provide the tooth model, the boundaries were segmented for the CAD modeling, and the tooth model automatically deformed and adjusted to produce a virtual design or reconstruction of the area and provide the output. Furthermore, all the included articles successfully reconstructed and modeled the restoration with a high degree of anatomic similarity compared with the original tooth. However, all the studies used a young dentition to

Table 5. Study characteristics of AI models for simulating and optimizing manufacturing casting process

Reference	AI Model	Application	Implementations	Performance
Mackert, ⁵⁰ 1988	Expert system	Diagnosing and suggesting solutions for casting related problems	ForCast (expert system) developed using expert system shell, available as user-supported software program (The Expert by S. Walton). Expert system asks users questions about defective casting and comes to conclusion about nature and cause of problem. It shows conclusion as well as suggested steps to correct problem, if possible, and prevent its reoccurrence. Knowledge base for system constructed using decision tree (inference networks) that derived from compilation of pertinent literature on casting failure analysis.	Not provided.
Matin et al, ²¹ 2017	Expert system (blackboard architecture, rule-based reasoning and iterative redesign method)	Computational modeling tools for cast design	Expert System with following functions: CAD modeling of simulation model for casting, fast modeling of gate design, CAD eligibility and cast ability check of model, estimation and running of program code for casting machine, and manufacturing time reduction of metal substructure. Development of simulation model: Scanned dental cast using IOS or laboratory scanner. ES creates user-defined feature (UDF) of scanned cast and converted into solid model using software program (Creo Parametric). Then, exported into numerical simulation for determination of casting parameters (640 default and 88 additional program codes with appropriate casting protocols). Blackboard architecture part of database connected to knowledge sources that contains data and hypotheses, with candidate partial solutions. Blackboard structured as weak hierarchy of levels. Levels structured in terms of set of casting parameters. Set of knowledge sources contained different kind of knowledge needed to solve problems and were modular, independent, and extensible. Assessment on 5 metal frameworks for metal-ceramic crowns. Dimensional deviation between designed and manufactured cast determined by using coordinate measuring machine (Zeiss Contura G2) and GOM Inspect software.	Dimensional deviation between simulation model of metal substructure and manufactured cast 0.018 mm. Roughness mean values measured on casting substructure varied from 1.935 μm to 2.778 μm. Manufacture of substructure showed that proposed ES allowed improvement of design process while reducing manufacturing time.

CAD, computer-aided design; CT, computed tomography; ES, expert system; IOS, intraoral scanner.

construct the tooth model, and limited studies considered the antagonist and contact points with the neighboring teeth on the automatization process. Developments are promising and have demonstrated the potential of AI models in CAD modeling for different designing features in prosthodontics. Further developments are needed to obtain a system able to learn different tooth geometries such as aged or worn dentition or different occlusion types and to produce a plausible reconstructed dental restoration for different clinical conditions.

Three studies³⁵⁻³⁷ analyzed the occlusal morphology similarity between the reconstructions made by an AI model integrated into a dental CAD software program for milling a restoration chairside and the original tooth geometry^{35,37} or the handmade reconstructions completed by dental laboratory technicians.³⁶ The results of the studies showed a successful AI model output obtaining a discrepancy of approximately 220 μm volume/area compared with the original tooth geometry^{35,37} or 310 μm volume/area compared with the handmade waxing performed by experienced dental laboratory technicians.³⁶ Comparisons among the studies are

difficult because of variations in the research protocols, such as the type of restoration simulated (inlay, onlay, or complete coverage restoration), manual retouches of the generic model suggested by the software program, or the measurement methods.

Only 1 included study developed an AI model (convolutional neural networks based on S-Octree structure) to automatically detect the finishing line of tooth preparations.¹⁹ The investigation used 380 digitized tooth preparation dies of premolar and molars for crowns from an unidentified source. The virtual dies were processed as the sparse point cloud with labels, marked with labels, and used to train a convolutional neural network model. The AI model was able to locate the margin line without manual interaction with an average accuracy ranging from 90.6% to 97.4%. However, the analysis of different variables, such as the type and depth of the finishing line and conicity, for the tooth preparation or the method used to prepare the virtual die were not described, which might have influenced the outcomes of the AI model. However, the study showed the potential of the automatization of finishing line location, which would

Table 6. Study characteristics of AI models for supporting design of removable partial dentures

Reference	AI Model	Application	Implementations	Performance
Maeda et al, ⁵¹ 1985	Expert system	Clinical decision support system for designing RPDs	Logic programming language Prolog-KABA (Kyoto Artificial Brain Associates) used with following features: (a) inference function with some logics, (b) back tracking, (c) pattern matching, (d) could be used on small computer, (e) graphic functions available, (f) programming possible with natural spoken language, (g) effective editing function. Database contained 3 main parts: - Data obtained from oral examinations, - Logic database extracted from textbooks and experienced prosthodontists' knowledge related to RPD design concept. - Contains examples of RPDs designs obtained from clinical treatments. Data of patients entered into expert system and suitable design for particular patient selected and shown as graphical displays with some comments.	Feasible execution of expert system for designing RPDs, generating graphic representation of design. Performance evaluation: Not provided.
Maeda et al, ⁵² 1987	Expert system	Clinical decision support system for designing RPDs	Data base constructed using books and expert opinions, clinical database collected from clinical examination, and graphic data of each partial denture component. Operator introduces information (patient's name, occlusion type, periodontal chart, tooth mobility, number and position of teeth missing and present, abutment teeth selection). Expert system provides support on decision making for selecting components of an RPD such as major and minor connector, or clasps.	Feasible execution of expert system for designing RPDs, generating graphic representation of design. Using databases within expert system made following feasible: - Expert system execution time faster than without databases - Design suggested by expert system more likely and efficiently lead to reasonable and rational answers - Denture designs for clinical treatments could be stored and readily used as references for practice and education. Performance evaluation: Not provided.
Beaumont and Bianco, ⁵³ 1989	Expert system (MacRPD)	Clinical decision support system for designing RPDs	Program presents interactive user interface that presents questions and requests answers based upon Kennedy classification system, to select major connectors for both dental arches. User provides numerical input with computer keyboard, prompted by questions that appear on computer screen, and then presented with graphic representation of specific partially edentulous situation. As program continues, suggestions for choice of direct retainer subsequently obtained by user with similar interactive interface. Software program concludes with graphic representation of design and laboratory prescription. Underlying decision tree structure (heuristic search) to guide design process.	Feasible execution of expert system for designing RPDs, generating graphic representation of design and laboratory prescription. Performance evaluation: Not provided.
Beaumont, ⁵⁴ 1989	Expert system (MacRPD)	Clinical decision support system for designing RPDs	Computer-based software program intended to aid in RPD design. Screen design based on dental laboratory work authorization, providing unifying theme. Instructional design: Program composed of approximately 160 RPD designs that may be viewed by user in free-form manner. Interactive user interface and branching in nature, thus allowing data search to be conducted in nonlinear fashion and completely at user discretion. Design data selected on basis of Kennedy classification, presence or absence of modification spaces, presence or absence of tori, and location of retentive undercut on abutment teeth. Software program concludes with graphic representation of design and laboratory prescription.	Feasible execution of expert system for designing RPDs, generating graphic representation of design and laboratory prescription. Performance evaluation: Not provided.
Wicks and Pennell, ⁵⁶ 1990	Expert system	Clinical decision support system for designing RPDs	Program with systematic approach to suggest design substantiated by principal concepts. User identifies missing teeth in cast graphic, indirect retention suggested if applicable, indicated major and minor connectors listed.	Feasible execution of expert system for designing RPDs, generating graphic representation of design and laboratory prescription. Performance evaluation: Not provided.

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Table 6. (Continued) Study characteristics of AI models for supporting design of removable partial dentures

Reference	AI Model	Application	Implementations	Performance
			User can access collection of text and enhanced illustrations at any stage of design process. Questions related to periodontal health, residual ridge, esthetics, and occlusal considerations located throughout program. Software program allows user's choice preferences. When process finished, graphic representation and laboratory prescription is generated.	
Hammond et al, ⁵⁷ 1993	Expert System	Clinical decision support system for designing RPDs	Implemented in logic programming language PROLOG on Macintosh II fx microcomputer. Authors also defined rules related to geometrical shape and positions of icons representing teeth and denture components. Database: Rules adopted from 2 textbooks, influenced by proceedings of international conference on RPDs (Bates et al, 1984; Davenport et al, 1988)	Feasible execution of prototype combining direct manipulation and critiquing. Performance evaluation: Not provided.
Hammond et al, ³⁸ 1993	Knowledge-based System (RaPiD)	Clinical decision support system for designing RPDs	Database: Design rules adopted obtained from atlas for design of RPDs influenced by proceedings of international conference. Software program: RaPiD software program with 3 components: graphical user interface, design knowledge base, and inference engine. RaPiD integrated CAD, knowledge-based systems and databases, and employed logic-based representation to unify them. User's manipulation of icons, that show developing design, interpreted as set of transactions on logic database of design components. Rules of expertise represented as constraints in logic and design alterations subjected to checking of these constraints. Critique presented to user when design rules contravened as result of some alteration. RaPiD evaluation: 22 RPDs designs for 22 patients. Design produced by RaPiD software program evaluated by clinician and dental technician treating patients using questionnaire (design diagram, design instructions, and patient description).	Evaluation of RaPiD design by clinician and dental technician for 22 patients using questionnaire: All but 3 of the 154 responses good or very good.
Fitzpatrick et al, ³⁹ 1994	Knowledge-based system (RaPiD)	Educational use for teaching design RPDs	Purpose: RaPiD software program for educational purposes with 2 modes: automatic and manual. In manual mode, 2 levels present: 'student' and 'expert'. In student mode, user who infringes on design rules immediately made aware of infringement allowing system to control user's progression through design process to obtain optimal solutions. RaPiD uses variety of critiquing strategies to alert users of their errors. Two of main strategies: critique issued when user completed proposed alteration and critique issues immediately upon user's radical misuse of tool.	Not provided.
Davenport and Hammond, ⁴⁰ 1996	Knowledge-based system (RaPiD)	Clinical decision support system for designing RPDs	Purpose: Validating RaPiD reliability by surveying prosthodontists. Database: literature review related to RPDs design that concluded with 119 design rules (denture base, rests, clasps, connectors). Survey performed from 119 design rules and consulted with 10 expert prosthodontists from the School of Dentistry in Birmingham. Consulted experts suggested another extra 6 design rules. Agreement or rejection for design rules based on 50% agreement among 10 prosthodontists. In this survey, total of 113 of 125 rules obtained agreement (90%). Revised questionnaire completed into another 16 dental schools and 5 faculty members from University of Birmingham. At Birmingham, focus groups sessions	RaPiD database implemented with new 98 design rules.

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Table 6. (Continued) Study characteristics of AI models for supporting design of removable partial dentures

Reference	AI Model	Application	Implementations	Performance
			performed to discuss design rules disagreements. Only 4 of 125 rules remained with no majority view for or against them. Result incorporated in national survey. Dental schools suggested another 14 design rules, those rules would be considered in future but not included in this study. 125 design rules surveyed; majority of schools agreed with 98 rules (78.4%). Those 98 rules considered to implement in RaPiD.	
Davenport et al, ⁴¹ 1996	Knowledge-based system (RaPiD)	Clinical decision support system for designing RPDs	Results for RPDs design rules revised by 17 schools and prosthodontists ⁵⁹ described in manuscript: 20 rules for denture bases, 21 for rests, 38 for clasps, 22 maxillary connectors, and 21 for mandibular connectors. 19 design rules included in RaPiD software program. Remaining rules not yet implemented grouped according to their perceived importance and in relation to anticipated difficulty with which they might be incorporated into system. Rules not yet implemented expected to cause most difficulty in terms of knowledge representation likely to be those involving most complex geometrical reasoning.	RPD design rules identified as part of comprehensive survey would be implemented in further development of RaPiD system. Need also existed for building versions of RaPiD to suit local requirements that did not conform to set of rules established as norm. Performance evaluation: Not provided.
Modgil et al, ⁴² 1997	Knowledge-based system (extension of RaPiD using computer vision and logic grammars)	Automatization of RPDs design	Extension RaPiD software program: - Added point distribution and active shape models (ASMs) to determine dentition from images of diagnostic casts. This enabled design to be customized to, and visualized against, image of patient's dentition. - Used path grammar, form of logic grammar, generating path linking ordered sequence of subcomponents. Using this, shape of important and complex prosthesis component can be automatically seeded. Training set: 54 maxillary and 61 mandibular casts photographed and used for arch width, tooth presence, and tooth position variation analysis. Dentition of patient represented, by set of polygonal tooth outlines comprising in total 444 vertices and referred to as template (similar to RaPiD). Each image in training set hand-annotated by manually moving each template point into place.	Extended version of RaPiD substantially automated design process right from photograph of dental cast to annotated and validated design diagram ready to guide manufacture. Performance evaluation: Not provided.
Chen et al, ⁴³ 2016	Knowledge-Based System (using ontology and case-based reasoning [CBR])	Automatization of RPDs design	CDSSinRPD (clinical decision support system) software program. - Input: Oral conditions of patient - Process box: Instantiation process transfers findings of oral examinations into structured terms, stored in ontology database, and processed by CBR tool. Ontology stores structured terms, relationships between them and situations represented by terms. CBR used to produce RPD design via mathematical models (cosine similarity) using standard situations in ontology. Database created from philosophy of Peking University Hospital and School of Stomatology and textbooks. - Output: RPD design in a text format. 104 partially edentulous patients selected randomly from patient database of Peking University Hospital and School of Stomatology. Chart and CDSSinRPD designs compared with their similarity values calculated by cosine similarity algorithm in descending order. Top 5 patients selected (similarity rank).	Precision at position 5 of retrieved treatments 0.67 and at top of curve 0.96. Mean average of precision (MAP) 0.61 and normalized discounted cumulative gain (NDCG) 0.74.

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Table 6. (Continued) Study characteristics of AI models for supporting design of removable partial dentures

Reference	AI Model	Application	Implementations	Performance
Chen et al, ⁴⁴ 2020	Knowledge-Based System	Automatization of RPDs design	Uses previously developed CDSSinRPD software program⁴³ to generate RPD design as output in textual format. Converts this textual design into 2D RPD diagrams. Workflow consists of 3 main steps: - Annotations and orientations of teeth established from base maps - Built-in rules incorporated to describe variations cause by interactions of RPD components. - Software program draws each component by using series of curve functions. Evaluation: 112 RPD laboratory prescriptions (52 in maxilla, 60 in mandible) randomly selected from database of Peking University Hospital and School of Stomatology. Designs transformed into structured texts as inputted software program before being evaluated. 2 prosthodontics faculty members evaluated output designs for 112 patients as "correctness," "need for improvement," or "error."	In maxilla, average correctness rate 0.75, improvement rate 0.23, and error rate 0.02. In mandible, average correctness rate 1.00. Precision value under different conditions calculated as measure of the software program's performance, in which value of precision calculated as number of correct diagrams divided by number of all diagrams. Average precision 0.75 in maxilla, 1.0 in mandible, and 0.88 in total. According to Kennedy classification, precisions were 1.00, 0.87, 0.79, and 0.88, respectively, from Kennedy I to IV.
Takahashi et al, ⁴⁹ 2021	Deep CNN	Classification of edentulous arches using Kennedy classification	Data: 1184 oral photographic images (748 maxillary and 438 mandibular) obtained from patients. Images consisted of 4 types of dental arches: edentulous, arches with posterior tooth loss, arches with anterior edentulous space, and intact dentition in each jaw. Images randomly divided into 2 datasets: training dataset (1016 images; 656 maxillary and 360 mandibular) testing dataset (168 images; 92 maxillary and 76 mandibular). Method: Deep CNN developed using Keras and TensorFlow. The ImageNet pretrained model of 152-layer residual network model (ResNet152) with fine tuning used for preprocessing, and dataset trained to classify type of dental arch. Training dataset separated into 10 batches for every epoch and 50 epochs run at learning rate of 0.01.	In maxilla: - Classification accuracy: 99.5% - Precision: 0.25 - Recall: 1.0 - F-measure: 0.4 - AUC: 0.99. In mandible: - Classification accuracy: 99.7% - Precision: 0.25 - Recall: 1.0 - F-measure: 0.4 - AUC: 0.98 Percentage of correct predictions of classifications was more than 95% for all types of dental arches.
Telford et al, ⁵⁵ 1989	Expert System (using CAL with VIPA)	Education of designing RPDs	Purpose: Combination of video and computer graphics for computer-aided learning in prosthetic dentistry. RPD design viewed as ideal vehicle for testing pedagogical approach, as well as underlying technology. Faculty could use system to produce or modify coursework with VIPA. CAL with VIPA: Authoring system for interactive video use. Starts with photograph of edentulous cast, followed by student identification of Kennedy classification. Denture bases superimposed into original cast's photograph. Similarly, tooth supporting, rests, and clasps designs identified using mouse. Lastly, connectors determined.	Video interaction for designing RPDs. Performance evaluation: Not provided.
Lechner et al, ⁵⁸ 1995	Expert System (rule-based system using CAL)	Education of designing RPDs	Purpose: Pilot program for CAL for teaching RPD design and facilitating students in mastering technical aspects of RPD design by leading students through sequence of steps from start to finish. Design philosophy conveyed by program followed that of Sydney University Dental School. Computer showed set of casts and then led students through design process by indicating answers using set of rules. Program would then question if rules correct for given example or if it needed modification. Program required users to give answers in variety of forms: click on specific areas of screen, yes/no answers, ticking multiple choice answers.	Video-interaction for designing RPDs. Performance evaluation: Not provided.

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Table 6. (Continued) Study characteristics of AI models for supporting design of removable partial dentures

Reference	AI Model	Application	Implementations	Performance
Lechner et al, ⁵⁹ 1998	Expert System (rule-based system using CAL)	Education of designing RPDs	Purpose: Combination of video and computer graphics provides computer aided learning in prosthetic dentistry. Graphic movement included as 3D effect, created by use of Quick-time VR, allowing students to click and drag photographs of diagnostic casts to quickly and clearly show any angle required. Control of timed answers promotes use of inquiry-based learning (IBL). Allowed program to be interactive and problem oriented. Design philosophy conveyed by program followed that of Sydney University Dental School.	Video-interaction for designing RPDs. Performance evaluation: Not provided.

2D, 2-dimensional; 3D, 3-dimensional; AUC, area under the curve; CAL, computer-aided learning; CNN, convolutional neural network; RPD, removable partial denture; VIPA, video interactive program author.

enhance the automatization of designing dental restorations using AI models integrated with dental CAD software programs.

Two included studies analyzed the potential of AI models on simulating and optimizing the manufacturing casting process of dental frameworks.^{21,50} In 1988, an expert system aimed to provide a supportive decision-making software for dental laboratory technicians on diagnosing and solving casting-related problems.⁵⁰ A detailed inference network was developed based on the literature knowledge at the time of the publication. The remaining study was able to apply an AI model to simulate and optimize the casting parameters for a cobalt-chromium²¹ crown framework fabrication. The results showed an improvement of the design process, minimizing the porosity on the cast metal and reducing the overall manufacturing time.

A unique study applied a BPNN to predict facial changes in edentulous patients about to be restored with a removable prosthesis.⁶² The AI software program aimed to assist in designing a virtual tooth arrangement and additively manufacturing complete dentures. A total of 48 participants (43 for AI training and 5 for AI testing) were scanned with and without the complete dentures by using a facial scanner. The BPNN was trained to obtain the relationship between facial elastic deformation and the feature template of the preoperative scan. The AI model was able to predict facial soft-tissue changes after the delivery of a complete denture. This AI application may allow an improved diagnosis and better communication with the patient. However, complete denture variables such as buccal flange thickness or occlusion type of the data set were not described, which might impact the accuracy of the prediction outcome.

A total of 17 articles evaluated the feasibility of AI models designing RPDs.^{38-44,49,51-59} From the 17 reviewed manuscripts, 10 studies described clinical decision support systems (expert systems) for designing

RPDs,^{38-41,51-54,56,57} 3 developed an automated RPD design AI model,⁴²⁻⁴⁴ 1 created an automated complete or partial edentulism classification software program,⁴⁹ and 3 developed computer-aided learning programs with interactive video for educational purposes.^{55,58,59} The main common challenge relies on the designing rules from which the software program is built up. Most of the reviewed studies are based on selected textbooks and expert opinions, which may represent a specific philosophy. This problem represents an example of the scientific lack of agreement regarding RPD design.²⁶⁻²⁸

Davenport and Hammond⁴⁰ and Davenport et al⁴¹ did an excellent job trying to survey designing rules at 17 dental schools (prosthodontic department) and with 70 prosthodontists.^{40,41} Although the agreement of a designing rule was considered with at least 50% of the experts in agreement on accepting or rejecting the design rule, which did not necessarily represent the majority of the surveyed people, the threshold might be adequate for this purpose. However, only British and Irish dental schools were consulted. Engaging international experts and dental school faculty would have made the system more acceptable globally. Nevertheless, the AI models developed might assist in forming a consensus on RPD design rules.

The 10 reviewed articles that developed or described clinical decision support systems for designing RPDs demonstrated a feasible AI application. Three studies developed and evaluated the performance of software programs providing an automated RPD design^{43,44} or categorizing completely and partially edentulous diagnostic casts by using the Kennedy classification for partial edentulism.⁴⁹

Chen et al⁴² developed an ontology and case-based reasoning software program for the automatization of RPD design. The designing rules were based on the philosophy of the Peking University Hospital and School of Stomatology. The output designs of RPDs for 104 patients were compared with those completed by

casts by using the Kennedy classification for partial edentulism.⁴⁹ The program was trained with 1016 images and tested with 168 images. The authors reported a percentage of correct predictions for complete and partially edentulous diagnostic cast images of more than 95% for all types of dental arches, the diagnostic accuracy being significantly higher in the mandibular than in the maxillary images. This application would represent the first step for designing an RPD.

Future research directions might consider the implementation of intraoral scans or digitized diagnostic casts to increase diagnostic accuracy. In terms of the massive data requirements that are common with deep learning methods, exploration of methods such as one-shot and less-than-one-shot learning methods may be an important step toward improving AI models for restorative dentistry. Because precise data interpretation is important in dental diagnosis, the standardization and benchmarking of data sets might facilitate the development of AI models.

CONCLUSIONS

Based on the findings of the present systematic review, the following conclusions were drawn:

1. AI models have shown the potential for providing a powerful diagnostic tool for tooth shade selection, automated restoration design, locating the preparation finishing line, optimizing the manufacturing casting, predicting facial changes in patients with removable prostheses, and designing RPDs, but they are still in development. Additional studies are needed to further develop and assess their clinical performance.
2. Based on the reviewed studies, 1 AI model developed for tooth shade selection obtained better shade matching compared with conventional visual selection methods; 14 studies reported the feasibility of automated design of dental restorations using different AI models; 1 AI model reviewed was able to locate the margin line without manual interaction, with an average accuracy ranging from 90.6% to 97.4%; 4 investigations developed AI algorithms that optimized the manufacturing casting process; 1 AI model was able to predict facial changes in patients using removable prostheses; and 17 studies demonstrated the feasibility of AI applications for designing RPDs.

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